IV B.TECH – I SEMESTER

Prompt Engineering

Skill Enhancement Course | JNTUA R23 · B.Tech CSE

	PROMPT ENGINEERING Skill Enhancement Course	L	T	P	C
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Course Objective:

This course delves into prompt engineering principles, strategies, and best practices, a crucial aspect in shaping AI models' behaviour and performance. Understanding Prompt Engineering is a comprehensive course designed to equip learners with the knowledge and skills to effectively generate and utilize prompts in natural language processing (NLP) and machine learning (ML) applications. This course delves into prompt engineering principles, strategies, and best practices, a crucial aspect in shaping AI models' behaviour and performance.

Course Out comes:

- Under standing the fundamentals and evolution of prompt engineering.
- Gaining the ability to craft effective closed-ended, open-ended, and role-based prompts.
- Learning to probe and stress-test AI models for bias and robustness.
- Applying prompt optimization techniques and performance evaluation methods.
- Mitigating bias and promoting ethical prompting practices in NLP/ML systems.

Module 1: Introduction to Prompt Engineering

- *Lesson 1: Foundations of Prompt Engineering*
- Overview of prompt engineering and its significance in NLP and ML.
- Historical context and evolution of prompt-based approaches.

Module 2: Types of Prompts and Their Applications

- *Lesson 2: Closed-Ended Prompts*
- Under standing and creating prompts for specific answers.
- Applications in question- answering systems.
- *Lesson 3: Open-Ended Prompts*
- Crafting prompts for creative responses.
- Applications in language generation models.

Module 3: Strategies for Effective Prompting

- *Lesson 4: Probing Prompts*
- Designing prompts to reveal model biases.
- Ethical considerations in using probing prompts.
- *Lesson 5: Adversarial Prompts*
- Creating prompts to stress-test models.
- Enhancing robustness through adversarial prompting.

Module 4: Fine-Tuning and Optimizing with Prompts

- *Lesson 6: Fine-Tuning Models with Prompts*
- Techniques for incorporating prompts during model training.
- Balancing prompt influence and generalization.
- *Lesson 7: Optimizing Prompt Selection*
- Methods for selecting optimal prompts for specific tasks.
- Customizing prompts based on model behavior.

Module 5: Evaluation and Bias Mitigation

- *Lesson 8: Evaluating Prompt Performance*
- Metrics and methodologies for assessing model performance with prompts.
- Interpreting and analyzing results.
- *Lesson 9: Bias Mitigation in Prompt Engineering*

- Strategies to identify and address biases introduced by prompts.
- Ensuring fairness and inclusivity in prompt-based models.

Module 6: Real-World Applications and Case Studies

- *Lesson 10: Case Studies in Prompt Engineering*
- *Exploration of successful implementations and challenges in real-world scenarios.*
- *Guest lectures from industry experts sharing their experiences.*

Text books:

- "Prompt Engineering in Action" – *Danny D. Sullivan*
- "The Art of Prompt Engineering with Chat GPT: A Hands-On Guide" – *Nathan Hunter.*

Reference Books:

- "Prompt Engineering in Practice" – *Michael F. Lewis*
- "Mastering AI Prompt Engineering: The Ultimate Guide for Chat GPT Users" – *Adriano Damiao*
- "Writing AI Prompts For Dummies" – *Stephanie Diamond and Jeffrey Allan*
- "Prompt Engineering Guide" (Online Resource) – *promptingguide.ai*

Online Resource link :

https://www.udemy.com/course/understanding-prompt-engineering/?couponCode=NVDINCTA35TRT

(Common to All Branches of Engineering)

R-23

Course Code	Gender Sensitization	L	T	P	C
		0	0	2	0
Pre-requisite		**Semester**			
Course Objectives:					
To enable students to understand the gender related issues, vulnerability of women and men To familiarize them about constitutional safeguard for gender equality To expose the students to debates on the politics and economics of work To help students reflect critically on gender violence To make them understand that gender identities and gender relations are part of culture as they shape the way daily life is lived in the family as well as wider community and the workplace.					
Course Outcomes (CO):					
COs Statements Blooms level CO1 Understand the basic concepts of gender and its related terminology L1, L2, **CO2** Identify the biological, sociological, psychological and legal aspects of gender. L1, L2 **CO3** Use the knowledge in understanding how gender discrimination					

works in our society and how to counter it. L3 **CO4** Analyzethe gendered division of labour and its relation to politics and economics. L4 **CO5** Appraise how gender-role beliefs and sharing behaviour are associated with more well-being in all culture and gender groups L5 **CO6** Develop students' sensibility with regard to issues of gender in contemporary India L3					

SYLLABUS

Unit-1UNDERSTANDING GENDER

Introduction: Definition of Gender-Basic Gender Concepts and Terminology-Exploring Attitudes towards Gender-Construction of Gender-Socialization: Making Women, Making Men - Preparing for Womanhood. Growing up Male. First lessons in Caste.

Unit-2 GENDER ROLES AND RELATIONS

Two or Many? -Struggles with Discrimination-Gender Roles and Relations-Types of Gender Roles- Gender Roles and Relationships Matrix-Missing Women-Sex Selection and its Consequences- Declining Sex Ratio- Demographic Consequences-Gender Spectrum -

Unit-3 GENDER AND LABOUR

Division and Valuation of Labour-Housework: The Invisible Labor- "My Mother doesn't Work." "Share the Load."-Work: Its Politics and Economics - Fact and Fiction- Unrecognized and Unaccounted work -Gender Development Issues-Gender, Governance and Sustainable Development-Gender and Human Rights-Gender and Mainstreaming

Unit-4 GENDER-BASED VIOLENCE

The Concept of Violence- Types of Gender-based Violence-Gender-based Violence from a Human Rights Perspective-Sexual Harassment - Domestic Violence - Different forms of violence against women - Causes of violence, impact of violence against women - Consequences of gender-based violence

Unit-5 GENDER AND CULTURE

Gender and Film-Gender and Electronic Media-Gender and Advertisement-Gender and Popular Literature- Gender Development Issues-Gender Issues-Gender Sensitive Language- Just Relationships

Prescribed Books

- A.Suneetha, Uma Bhrugubanda, et al. *Towards a World of Equals: A Bilingual Textbook on Gender*", Telugu Akademi, Telangana, 2015.
- Butler, Judith. *Gender Trouble**: Feminism and the Subversion of Identity***. UK Paperback Edn. March 1990

Reference Books

- Wtatt, Robin and Massood, Nazia, *Broken Mirrors: The dowr**y** Problems in India**,*London : Sage Publications, 2011
- Datt, R. and Kornberg, J.(eds), *Women in Developing Countries*, *Assessing Strategies for Empowerment*, London: Lynne Rienner Publishers, 2002
- Brush, Lisa D., *Gender and Governance*, New Delhi, Rawat Publication, 2007
- Singh, Direeti, *Women and Politics World Wide*, New Delhi, Axis Publications, 2010
- Raj Pal Singh, Anupama Sihag, *Gender Sensitization: Issues and Challenges* (English, Hardcover), Raj Publications, 2019
- A.Revathy& Murali, Nandini, *A Li**fe in Trans A**ctivism*(Lakshmi Narayan Tripathi). The University of Chicago Press, 2016

Online Resources:

- Understanding Gender

chrome-extension://kdpelmjpfafjppnhbloffcjpeomlnpah/https://www.arvindguptatoys.com/arvindgupta/kamla-gender1.pdf

https://onlinecourses.swayam2.ac.in/nou24_hs53/preview

- Gender Roles and Relations

https://www.plannedparenthood.org/learn/gender-identity/sex-gender-identity/what-are-gender-roles-and-stereotypes

https://www.verywellmind.com/understanding-gender-roles-and-their-effect-on-our-relationships-7499408

https://onlinecourses.swayam2.ac.in/cec23_hs29/preview

- Gender and Labour

https://www.economicsobservatory.com/what-explains-the-gender-division-of-labour-and-how-can-it-be-redressed

https://onlinecourses.nptel.ac.in/noc23_mg67/preview

- **GENDER-BASED VIOLENCE**

https://eige.europa.eu/gender-based-violence/what-is-gender-based-violence?language_content_entity=en

https://www.worldbank.org/en/topic/socialsustainability/brief/violence-against-women-and-girls

https://onlinecourses.swayam2.ac.in/nou25_ge38/preview

- **GENDER AND CULTURE**

https://gender.study/psychology-of-gender/culture-impact-gender-roles-identities/

https://sociology.iresearchnet.com/sociology-of-culture/gender-and-culture/

https://archive.nptel.ac.in/courses/109/106/109106136/

Abdulali Sohaila. "I Fought For My Life…and Won."Available online (at: http://www.thealternative.in/lifestyle/i-fought-for-my-lifeand-won-sohaila-abdulal/

OPEN ELECTIVES